

Quiz 5 - A

Q1: Define a relation **R** on **Z** as follows: For all integers m and n ,
$$m R n \Leftrightarrow 5 \mid (m^2 - n^2)$$

Is $(-8) R 2$? **(2 Marks)**

Q2: Determine whether the following relation is reflexive, symmetric, transitive:

O is the relation defined on **Z** (set of all integers) as follows: For every $m, n \in \mathbb{Z}$, $m O n \Leftrightarrow (m - n)$ is odd. **(4 Marks)**

Q3: Consider the following three relations defined on $A = \{1, 2, 3\}$:

- $R_1 = \{(1,1), (2,2), (3,3), (1,2), (2,1)\}$
- $R_2 = \{(1,2), (2,3), (1,3)\}$
- $R_3 = \{(1,1), (2,2), (1,2), (2,1), (3,3), (2,3), (3,2)\}$

(a) Which of the three relations is transitive but not symmetric? **(1 mark)**

(b) Which relation is symmetric but not transitive? **(1 mark)**

(c) Does R_3 have all 3 properties? Justify your answer by checking all three of them. **(2 marks)**